Journal of the Mechanics and Physics of Solids 140 (2020) 103945

Contents lists available at ScienceDirect

Journal of the Mechanics and Physics of Solids

journal homepage: www.elsevier.com/locate/jmps

Cross-scale characterization of the elasticity of shales: Statistical nanoindentation and data analytics

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ARTICLE INFO

Article history:
Received 11 June 2019
Revised 7 March 2020
Accepted 17 March 2020
Available online 20 March 2020

Keywords:
Data analytics
Nanoindentation
Shale
Surround effect
Young's modulus

ABSTRACT

Shales are a class of multiscale, multiphase, hybrid inorganic-organic composite materials exhibiting both frictional and cohesive behavior, and it is very challenging to characterize and interpret their complex mechanical properties. A statistical nanoindentation approach with pertinent viable data analytics was developed to probe the mechanical properties of shales across different length scales. Grid nanoindentation experiments with continuous stiffness measurement performed on shales to relatively large depths of 6–8 μm obtained massive data, which were processed by the new data analytics: segmentation at selected depths of a great number (e.g., >500) of continuous Young's modulus versus indentation depth curves obtained from unknown constituent phases yielded multiple discretized sub-datasets that were processed to extract individual phases' elastic moduli at respective segmentation depths via probability density function (PDF)-based deconvolution; these depth-dependent Young's moduli of each phase were then fitted by a newly proposed surround effect model, leading to determination of the properties of both individual phases at the nano/micro-scales (i.e., virtually infinitesimal depths) and the bulk rock at the macroscale (i.e., ~10–100 μm depths). A significant advantage of this massive data-based indentation approach is that the mechanical properties of composite materials such as shales can be probed across different scales by a single measurement technique. In addition, a new criterion, termed Bin Size Index, was formulated for selecting depth-dependent, rational, optimized bin sizes for PDF construction. For the studied shales, results show that five mechanically-distinct phases are discerned, including a virtual interface phase between hard and soft constituents accounting for a majority of indents. Coincidently, the Young's modulus of the bulk rock is nearly the same as that of the interface phase, suggesting that the macroscopic properties of similar composites may be estimated from measurements on the interface of two phases with contrasting mechanical properties. Finally, this approach can guide the selection of appropriate indentation depths to probe the mechanical properties of both highly heterogeneous bulk materials at the macroscale and their individual constituent phases at the nano/micro-scale.

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https://doi.org/10.1016/j.jmps.2020.103945
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1. Introduction

Crude oil and natural gas recovered from shales have become an important source of hydrocarbon energy since the economical feasibility of hydraulic fracturing was proven in the late 20th century (Andrews, 2006; Rao, 2012). Same as other reservoir formations (e.g., sandstones), the elasticity and other mechanical properties of shales such as strength and fracture toughness have vital importance for the safe, economical, efficient, and sustainable recovery of hydrocarbons therein. For instance, geophysical subsurface exploration for the search, discovery, and estimation of oil and gas reserves requires knowledge of the elasticity of the bulk rock as well as its constituents; the success of hydraulically fracturing tight formations is highly dependent upon fracture toughness, a mechanical parameter closely related to elastic modulus (Yasui et al., 2017; Zeng et al., 2019); a more recently recognized issue is that the rapid reduction of production rate in shales is often observed after the first 8–12 months of production, which may be attributed to the significant degradation in stiffness of shales (i.e., shale softening) caused by the physical interactions and chemical reactions between the rock and fracturing fluids during hydraulic fracturing and subsequent hydrocarbon extraction processes (Baihly et al., 2010; Wang, 2016). Therefore, the planning, design, construction, and operation of oil/gas recovery projects require more thorough knowledge of the mechanical properties of shales, which is a long-standing issue faced by the petroleum industry.

Shales, generally known as a type of fine-grained sedimentary rocks, are mainly composed of clay minerals and other secondary phases (e.g., quartz, calcite, and pyrite), as well as organic matter (e.g., kerogen). Owing to the wide size range of constituent particles (e.g., from <2 μm for clays to a few millimeters for sand particles) and different mineralogical compositions, shales are a class of complex, porous, multiphase, and multiscale composite materials. Microscopic observations via optical and electron microscopy or other imaging techniques show that fine-grained platy clay particles in shales exist in the form of highly compressed, preferentially oriented clay matrix that acts as the major load-supporting medium, while relatively coarse, anhedral, equidimensional silt and sand-sized particles are embedded as isolated solid inclusions within a mixed clay and organic matrix (the latter exists primarily in the voids of the clay matrix) (Hornby et al., 1994; Kanitpanyacharoen et al., 2012; Schieber and Baird, 2001; Seiphoori et al., 2017). Moreover, as a sedimentary rock, these constituent phases were initially highly compacted upon sedimentation and burial, followed by the formation of interparticle cementation via geological processes such as diagenesis, hydrothermal alterations, dissolution, and precipitation, among others, leading to the formation of rocks that exhibit both frictional and cohesive behavior. Therefore, the complex mechanical properties of shales are affected by not only the mineralogical composition, particle size, and fraction of each constituent, but also the porosity and interparticle cementation.

Understanding the mechanical properties of bulk shale and its constituent phases is of vital importance for the geophysical subsurface exploration and subsequent long-term production of hydrocarbons from shale formations. Prior work studying the diverse aspects of mechanical properties of shales at multiple length scales show that it is very challenging to characterize and interpret the mechanical behavior of shales because of their heterogeneity and anisotropy (Johnston and Christensen, 1995; Josh et al., 2012; Sondergeld and Rai, 2011; Sone and Zoback, 2013a, b; Vernik and Nur, 1992). Traditional measurement techniques (e.g., triaxial testing) that are developed for macroscopic, relatively large samples can only yield the mechanical properties of the bulk rock as a composite, but are not capable of probing or extracting the mechanical properties of tiny-sized individual constituents (e.g., clay, silt, and sand particles). Due to the fact that the same type of mineral may behave differently in different rock formations or even in the same type of rocks but with different geological settings or histories, macroscopic mechanical characterization is capable of neither identifying such differences nor directly determining the mechanical properties of individual constituents.

In the past three decades, nanoindentation testing has been extensively used to probe the small-scale mechanical properties (i.e., mostly Young's modulus and hardness) of a wide variety of materials (Bull, 2005; Chen et al., 2010; Finke et al., 2001; Franke et al., 2008; Hou et al., 2016; Menčík et al., 1997; Oliver et al., 2007; Oliver and Pharr, 2004; Song et al., 2017; Zhang et al., 2010), including shales and other rocks (Ahmadov et al., 2009; Bennett et al., 2015; Bobko and Ulm, 2008; Ulm et al., 2007; Ulm and Abousleiman, 2006; Veytskin et al., 2017; Zhu et al., 2007). Moreover, this non-destructive technique allows for multiple or even massive measurements from small-sized samples, and results are averaged or statistically analyzed over hundreds or even thousands of measurements to minimize random errors. Unlike traditional macroscale triaxial testing that usually requires certain size and geometry of samples trimmed from intact cores of at least a few inches in height, drill cuttings from well boring or post-testing rock fragments from a triaxial specimen, provided that they are not highly disturbed or cracked, can be used for nanoindentation testing.

More recently, a grid or statistical indentation approach has been developed to probe the mechanical properties of highly heterogeneous, multiphase materials, particularly those consisting of a more homogeneous matrix with embedded granular solid inclusions, such as shales and concretes, via a massive number of measurements (Constantinides et al., 2006). Further statistical deconvolution of the voluminous experimental data yields insights into the mechanical behavior of individual constituent phases, but not the bulk rock whose properties are typically obtained by different macroscopic testing. In addition, because different constituents in a shale usually have varied particle sizes, the maximum indentation depth selected for measuring the elasticity of individual constituents should not be fixed, but vary with different indent locations (e.g., based on the 10% rule of thumb, the maximum indentation depth should be limited to the 10% of the thickness or size of the tested material). Therefore, measurements obtained at a fixed indentation depth for all different constituents may yield misleading results unless the indentation depth is nominally infinitesimal. Finally, given the wide range of mineral constituents usually having unknown particle sizes, it is also a difficult task to select appropriate, variable indentation

Table 1
The studied shale samples recovered from the Longmaxi formation.

Sample ID	Sample tag	Well ID	Well depth (m)	Formation
A	1-42/46 (2)	YS108	2390.11–2390.38	S_1l
B	7-31/50 (2)	YS108	2506.40–2506.61	S_1l-O_3w

depths to balance the surface roughness effect (i.e., the depth should not be too small to overcome the roughness effect) and the 10% rule of maximum depth (i.e., the depth should not be too large to induce the effects of nearby surrounding phases). In summary, developing a viable technique to probe the mechanical properties of both individual phases and the bulk rock without the need of selecting an appropriate indentation depth is of great interest, particularly for testing highly heterogeneous materials such as shales.

The above statistical nanoindentation techniques bring an additional advantage for the study of composites such as shales. With the determination of the mechanical properties as well as fractions of individual constituent phases, the micromechanical models developed based on different homogenization and upscaling schemes can be used to estimate the pertinent macroscale properties of the bulk composites. In particular, the elastic modulus of a multiphase composite can be modeled by the Voigt and Reuss models for its upper and lower bounds, respectively (Hill, 1952; Reuss, 1929; Voigt, 1889). In addition, the VRH model has been developed to take the average of the upper and lower bounds (Mavko et al., 2009). Sone and Zoback (2013a) studied the macroscale elasticity and mineralogical composition of a range of shales with varying clay contents and their results were compared with the Voigt and Reuss models. Another widely used model is the Mori-Tanaka model that considers the microscale average matrix stress and local fluctuating stress due to the secondary hardened phase dispersed in the primary matrix (Mori and Tanaka, 1973). Although this model can provide a direct estimation of the macroscale bulk properties, it yet has certain limitations, such as the suitability for composites with low volume fractions of inclusions (Klusemann et al., 2012). In fact, other models developed by mean-field homogenization schemes such as Double Inclusion Method (Hori and Nemat-Nasser, 1993), Interpolative Double Inclusion Model (Pierard et al., 2004), and Self Consistent Schemes (Christensen and Lo, 1979; Hill, 1965) have their own limitations, leading to certain errors in the predicted results. Therefore, validation of some specific models and assessment of the model-predicted results require experimental data that would be ideally obtained from the same samples used to determine the individual constituents' properties.

This paper presents a newly developed massive data-based nanoindentation approach for characterizing the mechanical properties of shales, a multiscale, multiphase composite material with solid inclusions. Combined with the newly formulated, rational data analytics as well as a surround effect model, the elasticity of both individual phases and the bulk rock crossing different length scales can be determined via a single type of mechanical testing (i.e., nanoindentation). In general, probing the elasticity of individual phases requires nanoscale measurements with infinitesimal indentation depths, while determination of the bulk rock's mechanical properties requires measurements at very large depths. This approach offers many advantages over other conventional measurement methods, one of which is obtaining the cross-scale mechanical properties via a single type of mechanical testing. It can be expectedly implemented as an economical, effective, and efficient toolkit for cross-scale mechanical testing of rocks and other composite materials.

2. Materials and methods

2.1. Samples

The two studied shale samples, identified as Samples A and B, were recovered from a deep well drilled into the Longmaxi Formation, Sichuan basin, Southwest China at depths of 2390 and 2506 m, respectively (Table 1). Two types of samples were acquired (Fig. 1a): (1) post-failure fractured fragments from intact core samples previously subjected to triaxial compression testing, and (2) relatively large rock chips sifted out of drill cuttings. While the former was used for nanoindentation testing, the latter was solely used for compositional analysis via X-ray powder diffraction (XRPD). Particularly noteworthy is that the bedding plane orientation in the core fragments was always traced at various stages of sample preparation.

2.2. X-ray powder diffraction (XRPD)

The aforementioned drill cuttings were first crushed in percussion mortar and pestle to have the entire material pass through a 0.4 mm sieve, followed by wet-grinding with propan-2-ol (C_3H_8O) in a McCrone micronizing mill (The McCrone Group, Westmont, IL, USA) for 10 min, which usually produced a slurry consisting of particles of <4 μm in size (Locock et al., 2012). After drying, the pulverized sample powder was thoroughly and homogeneously mixed with 10 wt.% zincite (ZnO) used as an internal standard for quantitative XRPD analysis. The dry powder mixture was then mounted into the sample holder via the razor-tamped surface (RTS) method (Zhang et al., 2003) to prepare a powder mount with highly random orientation of platy clay particles. All XRPD measurements were conducted in a PANalytical X'Pert PRO X-ray diffractometer (Almelo, Netherlands) using Cu Kα radiation (with a wavelength of $\lambda = 1.5418$ Å) generated at 45 kV and 40 mA, a continuous scan range of 2°–64° 2θ, and a scan speed of 1° 2θ/min. The acquired diffraction patterns were analyzed by Profex,

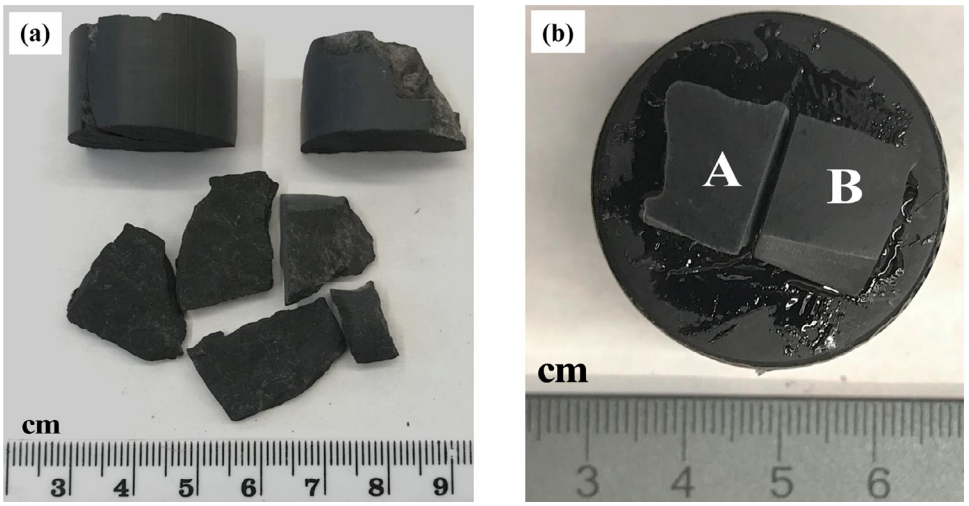


Fig. 1. (a) Two types of shale samples: rock core fragments (top) for nanoindentation, and drill cuttings (bottom) for X-ray powder diffraction; (b) The polished samples (A and B) glued on the aluminum puck for nanoindentation testing.

an open-source program with the Rietveld refinement (Doebelin, 2018; Doebelin and Kleeberg, 2015), to positively identify various mineral phases and estimate their fractions (i.e., both qualitative and quantitative analyses).

2.3. Statistical nanoindentation

Relatively large core fragments from fractured triaxial specimens with known bedding plane orientation were selected for nanoindentation testing. To prepare flat and smooth surfaces for indentation, selected core fragments were cut into small cuboids of ~15 × 10 × 10 mm with the base parallel to the bedding direction. The cuboids were then glued onto cylindrical aluminum pucks heated on a hotplate to 130 °C using the Crystalbond 509 amber resin (Aremco Products, Inc., New York, NY, USA) that melts at 130 °C and hardens upon cooling. Once the resin cooled down and hardened, the base surface was successively polished in a MetaServ 250 polisher (Buehler Inc., Lake Bluff, IL, USA) using abrasive papers with decreasing grit sizes (i.e., from #P180 to #P4000 designated by the Federation of European Producers of Abrasives (FEPA)) to attain a highly smooth and flat surface (Fig. 1b).

Nanoindentation testing was conducted on the polished surfaces of the shale samples under the continuous stiffness measurement (CSM) mode in a Keysight G200 nanoindenter (Keysight Technologies, Inc., Santa Rosa, CA, USA) equipped with a pre-calibrated, custom-made Berkovich diamond tip of <20 nm in radius and of ~25 μm in effective depth. For each sample, a total of 539 indents were made on the polished surface via 11 batches, each of which consisted of a randomly located 7 × 7 matrix with a spacing of 200 μm between two adjacent indents to prevent negative interference from each other (Sudharshan Phani and Oliver, 2019). All indents were set to a targeted maximum depth of ~8 μm, and load was increased at a constant strain rate ($\dot{h}/h$) of 0.05 1/s under an allowed thermal drift rate of 0.05 nm/s. The samples were allowed to creep for 60 s at the maximum load before unloading. The harmonic oscillation of the CSM mode was controlled by displacement with an amplitude of 2 nm and a frequency of 45 Hz.

In addition to aforementioned grid indents at random locations, two other sets of indentation testing were also conducted: (1) indenting directly on the boundaries between solid inclusions and clay matrix, and (2) ex-situ indentation on sand-sized individual quartz and carbonate particles. The former involved intentional measurements of the mechanical response of the solid inclusion-clay matrix interface, and a total of 20 indents individually selected under the built-in optical microscope were made, with the purpose of understanding indentation response of the interface between hard and soft phases. The latter was to obtain the ex-situ Young's moduli of pure quartz and carbonate sand particles of 150–250 μm in size, which were directly glued onto the cylindrical aluminum puck and then polished in the same way as shale samples. A total of ~30 indents were made on each type of sand particles.

For each indent, the recorded load versus indentation depth curves were analyzed by the Oliver-Pharr method (Oliver and Pharr, 1992). In particular, CSM-based measurements enabled continuous determination of the mechanical properties (e.g., Young's modulus) over the entire indentation depth (i.e., ~8000 nm in this study), because the contact stiffness S was determined based on the analysis of the dynamic response of the entire indenter-sample system to a harmonic vibration. For all data analyses, the Young's modulus and Poisson's ratio of the diamond indenter were 1141 GPa and 0.07, respectively, while a constant Poisson's ratio of 0.2 was assumed for all shale samples. In fact, prior work demonstrated that the error in the assumed Poisson's ratio of the sample has little influence on the estimation of its Young's modulus (Mesarovic and Fleck, 1999; Oliver and Pharr, 1992; Saha and Nix, 2002; Zhang et al., 2009).

Table 2
Qualitative and quantitative mineral compositions of the two shale samples determined by XRPD.

Sample ID	Mineral name and fraction (wt.%)							
	Quartz	Calcite	Dolomite	Pyrite	Albite	Muscovite	Chlorite	Illite
A	20	27	3	1	4	8	9	28
B	18	24	3	1	2	14	7	31

The theoretical basis for statistical indentation is that, for a heterogeneous material, the mechanical property (e.g., Young's modulus) of each phase is assumed to follow a Gaussian distribution. If a massive grid of indentations are performed on a heterogeneous material that consists of n different phases, the overall distribution of the experimentally obtained data can be treated as a combined response of n different Gaussian random variables. Hence, the mechanical properties of the constituent phases, including both solid inclusions and matrix, can be determined in-situ and distinguished from each other by implementing the probability density function (PDF) deconvolution, provided that the selected indentation depth or size is nominally very small or infinitesimal. In general, results obtained from the statistical PDF deconvolution are not necessarily unique when the initial estimates for the fitting process are different. Therefore, in this study, multiple sets of randomly initialized parameters were tried, and the final solution was selected by evaluating the fitting results with the combination of the mathematical perspective (i.e., minimizing the fitting errors) and rationality in mechanical testing. In addition, as discussed later, a new criterion for optimizing the number of intervals (m) or bin size to construct the experimental histograms was formulated to reduce the uncertainty of PDF-based deconvolution. Further details on PDF-based statistical deconvolution can be found in the literature (e.g., Constantinides et al., 2006; Ulm and Abousleiman, 2006).

2.4. Surface characterization

Three different techniques were employed to characterize the sample surface before and after indentation testing. The first one was the built-in optical microscope with the G200 nanoindenter. Observing and imaging sample surfaces prior to and after indentation testing used this technique to survey the surface, select interested zones and spots (e.g., the solid inclusion-clay matrix interfaces) for indenting, and examine the residual imprints.

The second technique, atomic force microscopy (AFM), was used to obtain the surface roughness. Upon polishing, the surface of Sample A was scanned under the contact mode using an MFP-3D AFM (Asylum Research, Santa Barbara, CA, USA) equipped with a NSC14 silicon probe (MikroMasch, Bulgaria) with a nominal tip radius, stiffness, and resonant frequency of 10 nm, 5 N/m, and 160 kHz, respectively. Before imaging, the resonant frequency and stiffness of the AFM cantilever were calibrated, and the shape of the probe was examined by imaging a TGT1 grating (NT-MDT, Zelenograd, Russia) to ensure that the tip of the probe was free of defects. The root-mean-square (RMS) roughness R_q of the imaged surface was determined by averaging data from 7 randomly selected zones, each with an area of $90 \times 90\ \mu m^2$.

The third technique was scanning electron microscopy (SEM), which was used to image the residual imprints after (but not before) indentation testing, in order to prevent the potential surface damage or disturbance caused by the energized electron beam. The indented surface was imaged in a FEI Magellan 400 XHR SEM (Thermo Fisher Scientific, Inc. Hillsboro, OR, USA) operated at a 5 kV beam voltage and 13 pA beam current, with a particular focus on the residual indents. No conductive coating was applied to the examined sample, since the electron beam was generated at a low voltage and very small current.

3. Results and discussion

3.1. Mineralogical composition

Fig. 2 shows the XRPD patterns of the two studied shale samples, while the quantitative results are summarized in Table 2. Obviously, there is not too much difference in the two XRPD patterns, which should be expected for two samples recovered from two nearby strata in the same wellbore. Three clay minerals, muscovite, chlorite, and illite, are present in both samples with a total clay fraction of 45 and 52 wt.%, respectively. Both samples contain significant amounts of carbonates, mainly calcite at 27 and 24 wt.% respectively, which are the major cementation agents contributing to the solid interparticle bonding. Quartz is the third major phase in the rocks with a fraction of 20 and 18 wt.%, respectively. Two minor phases with much smaller fractions are pyrite and albite, which are not unusually present in shales. With the quantitative results, it can be concluded that both samples are clay-rich shales with carbonates as the cementation agent and carbonates and quartz as the major solid inclusions.

3.2. Surface characteristics

As an example, Fig. 3 shows the micrographs of the polished surface of Sample A obtained by optical microscopy, SEM, and AFM. A general feature is that the surface is still not perfectly smooth, even after successive fine polishing, most likely

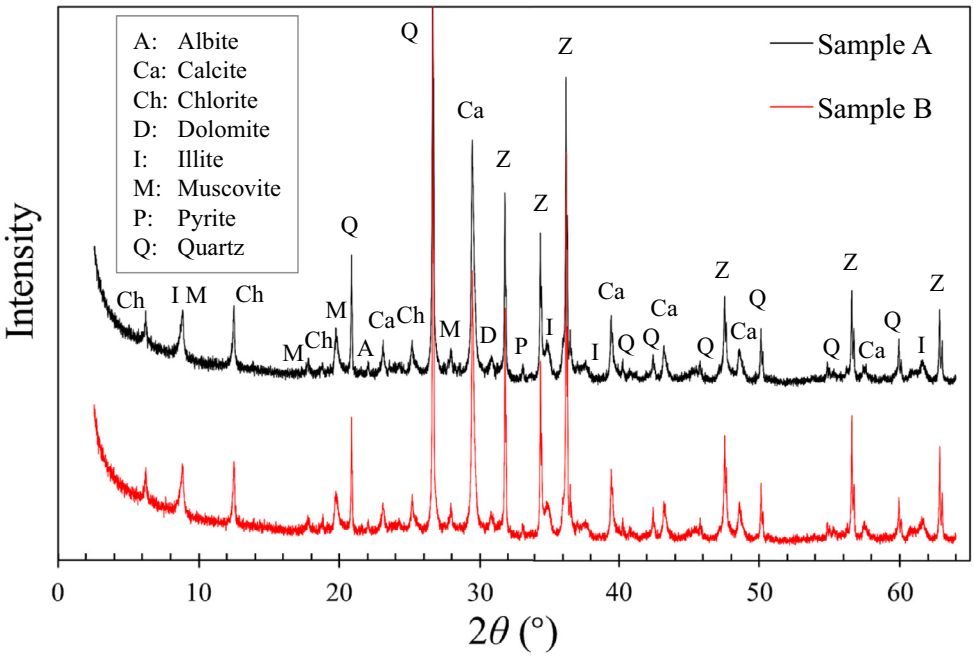


Fig. 2. XRPD patterns of the two shale samples (Note: Z stands for zincite, an added internal standard for quantitative analysis).

due to the rocks' multiple phases with varied hardness and stiffness. In fact, the optical micrographs (Fig. 3a and b) clearly show the contrasting, different mineral phases in the rock. For instance, the small shining spots are pyrite, while the clay matrix generally has a mottled grey-black pattern. Moreover, relatively large, monolithic particles can be identified from all of these images, and their sizes range from ~10 to >50 μm. As such, selective indentation on the particle-clay matrix interface (or the boundaries of these solid inclusions) becomes feasible. Overall, the monolithic, hard, sand-sized particles (i.e., including quartz, calcite, and pyrite) are randomly embedded in the clay matrix. Finally, quantitative AFM height images (Fig. 3e and f) enable the calculation of surface roughness as well as the observation of pores within the clay matrix. The RMS roughness R_q is 138 nm, which may affect the indentation measurements obtained at depths of <200 nm. These height images also show the pores with negative height values, which may host organic matter (e.g., kerogen) or be filled with dusts or debris from surface polishing.

3.3. Typical indentation behavior

A total of 539 indents was made on each of the two samples, resulting in 539 × 2 independent load-depth (*F-h*) and corresponding Young's modulus-depth (*E-h*) curves. It is infeasible and impractical to show all of these curves in this paper. As examples, Fig. 4a and b show 49 *F-h* and corresponding *E-h* curves from a randomly selected 7 × 7 indent matrix on Sample A. Among these *F-h* curves, the achieved maximum indentation depth varies from ~3 to 8 μm, and most curves terminate at h = ~5 μm, which is less than the preset targeted maximum depth of 8 μm. The reason is that the instrument's maximum load capacity is 600 mN, which limits the achievable maximum indentation depth, depending upon the mechanical properties (i.e., stiffness and hardness) of the indented materials. As such, subsequent data analysis just focuses on the data in the depth range of 0 to 5 μm, because only a few curves extend beyond h = 5 μm, resulting in insufficient number of data for further meaningful statistical analysis.

In Fig. 4b, the dependence of Young's modulus upon indentation depth is clearly observed in all curves. While some curves show that the directly measured, apparent E decreases with depth, others exhibit the opposite trend. Moreover, despite the significant serrations on the majority of these curves, the measured E among different curves varies significantly within the range of approximately 0 to 120 GPa at depths of <0.25 μm. However, as the depth increases to 4 μm, these curves gradually converge to a much smaller range. Noteworthy is that some curves show a delayed loading (Fig. 4a–c), indicated by the nearly zero load over a large range of depths (i.e., 2–3 μm) at the initial stage, and hence result in a relatively larger maximum indentation depth. A similar phenomenon was also observed for indentation on cement pastes (Hu et al., 2016). Such abnormal loading curves are caused either by the instable contact between the indenter tip and sample surface or by the incorrect surface detection when the indent is made on pores and small asperities on the surface. As illustrated in Fig. 4d, the indenter may incorrectly detect the surface of a pore loosely filled with polishing debris or dust and starts to record the data at Location 1, while the majority of the indenter tip is still not yet in contact with the solid

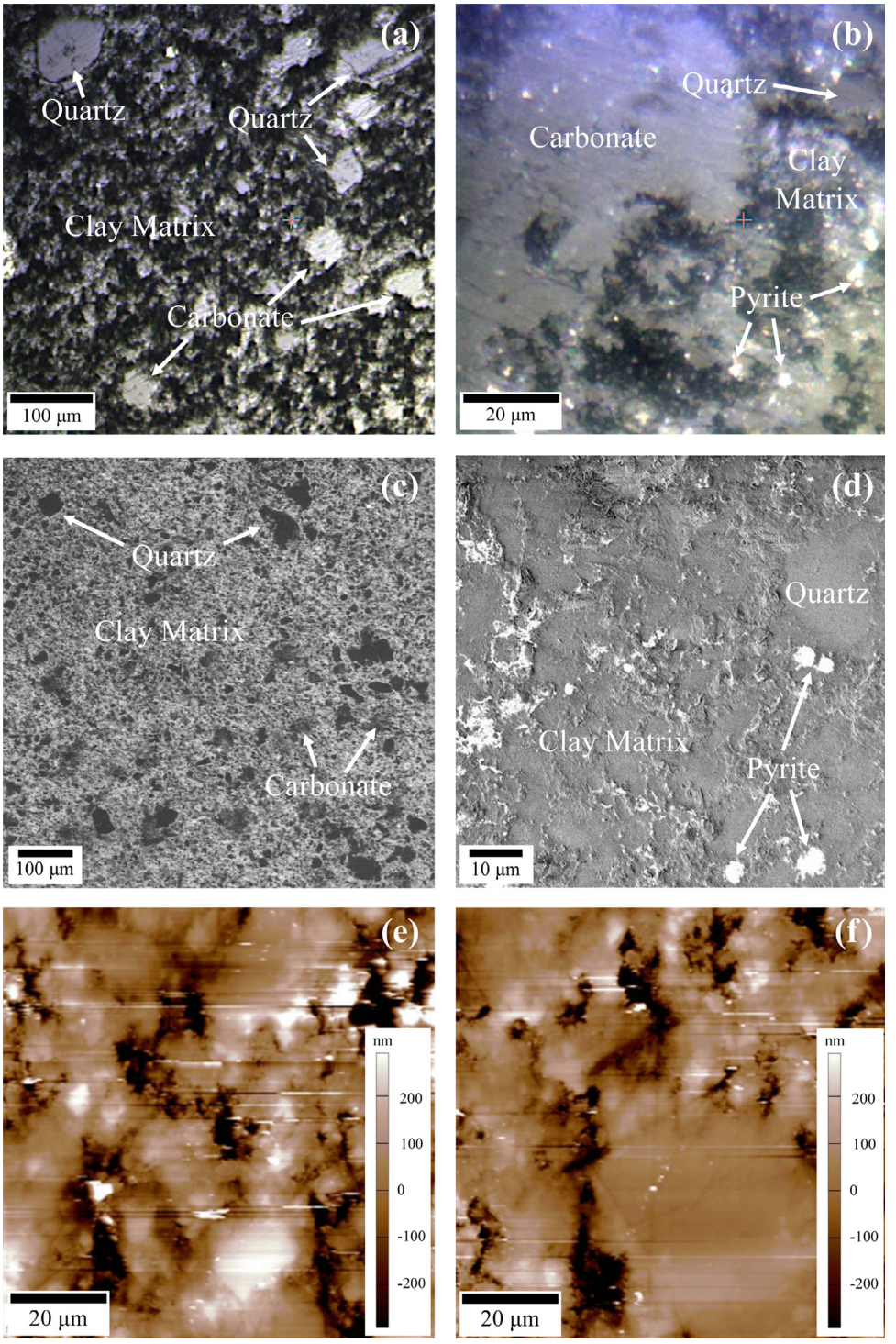


Fig. 3. Surface characteristics of Sample A: (a–b) optical micrographs showing the granular inclusions are embedded in the clay matrix; (c–d) SEM micrographs showing similar features; (e–f) AFM micrographs showing surface topography.

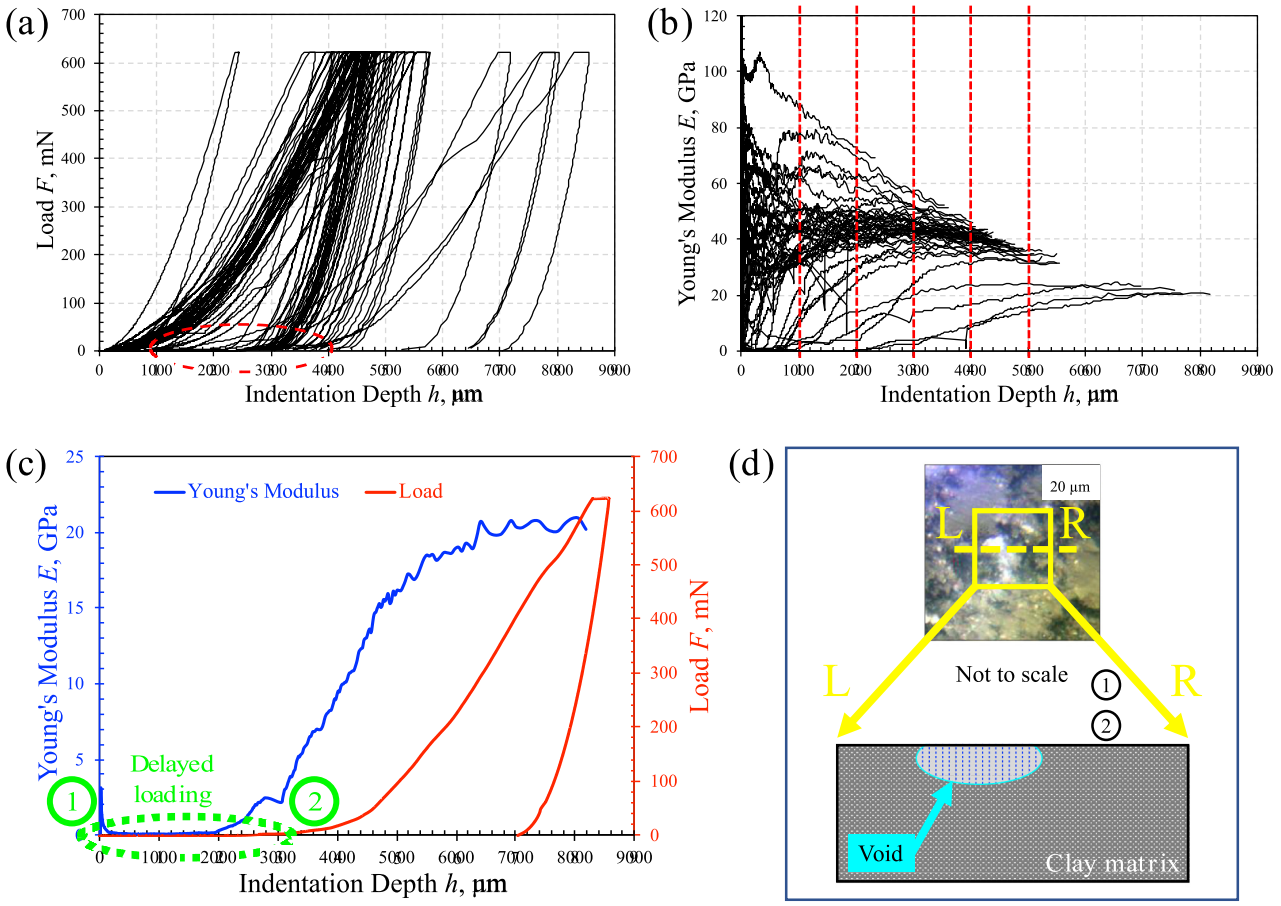


Fig. 4. Example indentation results from Sample A: (a) 49 indentation load versus depth curves; (b) corresponding 49 Young's modulus versus depth curves; (c) delayed loading phenomenon at some indents; (d) schematic illustration of the delayed loading when indenting at a void (Note: the two depths labeled as 1 and 2 in (d) correspond to the two points marked in the *F-h* curve in (c)).

sample until the indenter proceeds deeper to Location 2. Therefore, these curves with incorrect or abnormal detection of surface contact are excluded from subsequent statistical analysis.

3.4. Proposed data analytics

Since the CSM loading mode employing a harmonic displacement vibration of 2 nm in amplitude yields continuous measurements of Young's modulus over the entire depth, each curve with a maximum h of 5 μm can nominally yield 5000 independent data points. As such, a total of 500 curves contains 2.5×10^6 data points, which may be large enough to be considered as "big data". Moreover, the massive data-based approach also refers to the new data analytics proposed here to process such a large number of *E-h* curves, leading to the extraction of cross-scale elasticity of the shale samples.

As shown in Fig. 4b, no regular pattern can be observed in the *E-h* curves. In fact, this holds true for all *E-h* curves obtained from both samples. In other words, the unprocessed, directly measured *E-h* curves do not offer any meaningful insights into the complex, multiphase, and multiscale nature of the rocks. Although a simple arithmetic average over the entire spectrum of curves seems attractive for the determination of elastic modulus of the rock as a composite, it does not have a sound basis or misses the information on its multiple mineral phases and their mechanical properties at different length scales. In this paper, a new method of data analytics is proposed to process such an enormous volume of data that appear erratic and chaotic. This rational data analytics consists of a series of data processing and analysis summarized below and hence estimate the properties of mechanically distinct phases and the bulk rock:

- Data segmentation and extraction;
- Histogram construction and PDF deconvolution;
- Data re-assembly or re-integration;
- Surround effect modeling;
- Property determination.

The first step, data segmentation and extraction, is to segment the ~500 continuous *E-h* curves at selected indentation depths, hereafter termed segmentation depths (e.g., 5 depths with a 1.0 μm interval, as shown in Fig. 4b), resulting in multiple sub-datasets, each of which is extracted at the same fixed segmentation depth. In other words, each sub-dataset contains all the measured Young's moduli at a given segmentation depth, and these data then form a possible range of Young's modulus for all but unknown number of mineral phases in the rock, equivalent to the same number of random variables. In fact, for each of the two samples, a total of 16 sub-datasets, each of which contains ~539 measured Young's moduli, are created from all of the continuous *E-h* curves at 16 segmentation depths. In the second step, each sub-dataset making up a set of measurements of unknown number of random variables is then statistically analyzed by deconvoluting the overall PDF, whose construction requires the selection of an appropriate bin size for generating the experimental histograms. In this paper, variable but not constant, depth-dependent bin sizes were used for different sub-datasets extracted at different segmentation depths, depending upon the range of the measured Young's modulus values. Further details on how to select an appropriate bin size are given in the next section. Such deconvolution yields insightful information on the samples, including the number, property, and area or volumetric fraction of all mechanically distinct phases in the rocks. The third step is to re-assemble the discretized deconvoluted results by replotting the Young's modulus of the same phase against the segmentation depth. The re-integration of the discrete Young's moduli from the same phase but different segmentation depths enables the observation of depth-dependent, progressive transition of Young's modulus from individual phases at nano/micro-scale to the bulk rock at the macroscale. Such re-integration requires the identification of the virtual "interface phase" that serves as basis for this operation, as discussed later. In the final two steps, the re-integrated depth-dependent Young's moduli are fitted by a newly proposed surround effect model for each mechanically distinct phase. As discussed later, the model is so powerful that it yields an array of insightful information for the tested rock.

With the aid of newly developed data analytics, the ~539 erratic and chaotic *E-h* curves with mechanical contributions from different but unknown mineral phases are converted into only a few (e.g., 4–5) *E-h* curves, each of which describes the depth-dependent transition of the Young's modulus of a specific phase from truly nano/micro-scale to macroscale (i.e., cross-scale transition). As described in the following sections, the proposed viable data analytics enables determination of the mechanical properties of each individual mineral as well as the bulk rock, and also yields many useful insights into the characterization and understanding of the mechanical properties of shales and other similar complex materials.

3.5. Bin size selection

Upon segmentation of the *E-h* curves (Fig. 4b), generation of the experimental PDF (i.e., histogram construction) of the extracted sub-dataset (i.e., ~539 Young's modulus values at a fixed segmentation depth) for subsequent statistical deconvolution requires the selection of a proper and rational bin size (or bin width), which manifests the trade-off between creating a plot with too much (i.e., undersmoothing) and too little (i.e., oversmoothing) details with respect to the true distribution of one or more random variables (Wand, 1997). For instance, larger bin sizes (i.e., larger intervals) help reduce the noises induced from measurement randomness, whereas smaller bin sizes give greater precision to estimation of the true distribution. The experimental PDF with a smaller bin size usually requires more phases to capture more local features (Trtik et al., 2009), and the reasonable number of phases used to deconvolute the experimental PDF is chosen partially based on prior knowledge of the mineralogical composition of the sample, for example, determined by XRPD. However, the results of XRPD does not necessarily reflect the true number of mechanically distinct phases in shales, due to either too low volumetric fractions or too small particle sizes of some phases, among others.

Although much attempt has been made by theoreticians or statisticians to estimate the optimal bin size (Freedman and Diaconis, 1981; Izenman, 1991; Knuth, 2006; Pan and Politis, 2016; Scott, 1979; Shimazaki and Shinomoto, 2007; Taylor, 1987; Terrell, 1990; Terrell and Scott, 1985), its selection is yet mostly empirical and a matter of individual's choice. In this paper, a new criterion is proposed to optimize a rational bin size for a given sub-dataset that consists of measurements of multiple independent random variables (i.e., multiple phases of the rock in this paper). It is logical to believe that the bin size should generally depend upon the range (i.e., the difference between the maximum and minimum values) of the processed sub-dataset as well as the number of independent random variables. This new criterion is based on a new concept, termed Bin Size Index (BSI), which is formulated as follows in detail.

The sub-dataset from Sample A extracted at segmentation depth of 0.5 μm is used as an example. First, the Freedman-Diaconis rule (Freedman and Diaconis, 1981) was adapted to provide an estimate of the initial bin size b:

$$b = \frac{2 \times \mathrm{IQR}(x)}{n^{1/3}} \tag{1}$$

where x is the original measurements of Young's modulus extracted at this depth, IQR is the interquartile range of these measurements, and n is the total number of these measurements. As such, an initial $b = 5.3$ GPa was determined from above equation for the Young's moduli obtained at $h = 0.5$ μm. Concerns with the significant effect of the surface roughness on the data quality at extremely shallow depths, as discussed later, prevent the use of sub-datasets extracted at other smaller segmentation depths (e.g., 0.25 μm). Thus, the reason for selecting the starting depth of 0.5 μm instead of other smaller ones is that the results obtained at smaller depths are more significantly affected by surface roughness and surface contamination.

Then five different trial bin sizes, $b = 1.0$, 2.0, 3.0, 4.0, and 5.0 GPa, are used to construct five PDF histograms, followed by PDF deconvolution with number of phases ranging from four to eight (Fig. 5). For each deconvolution result pertinent to

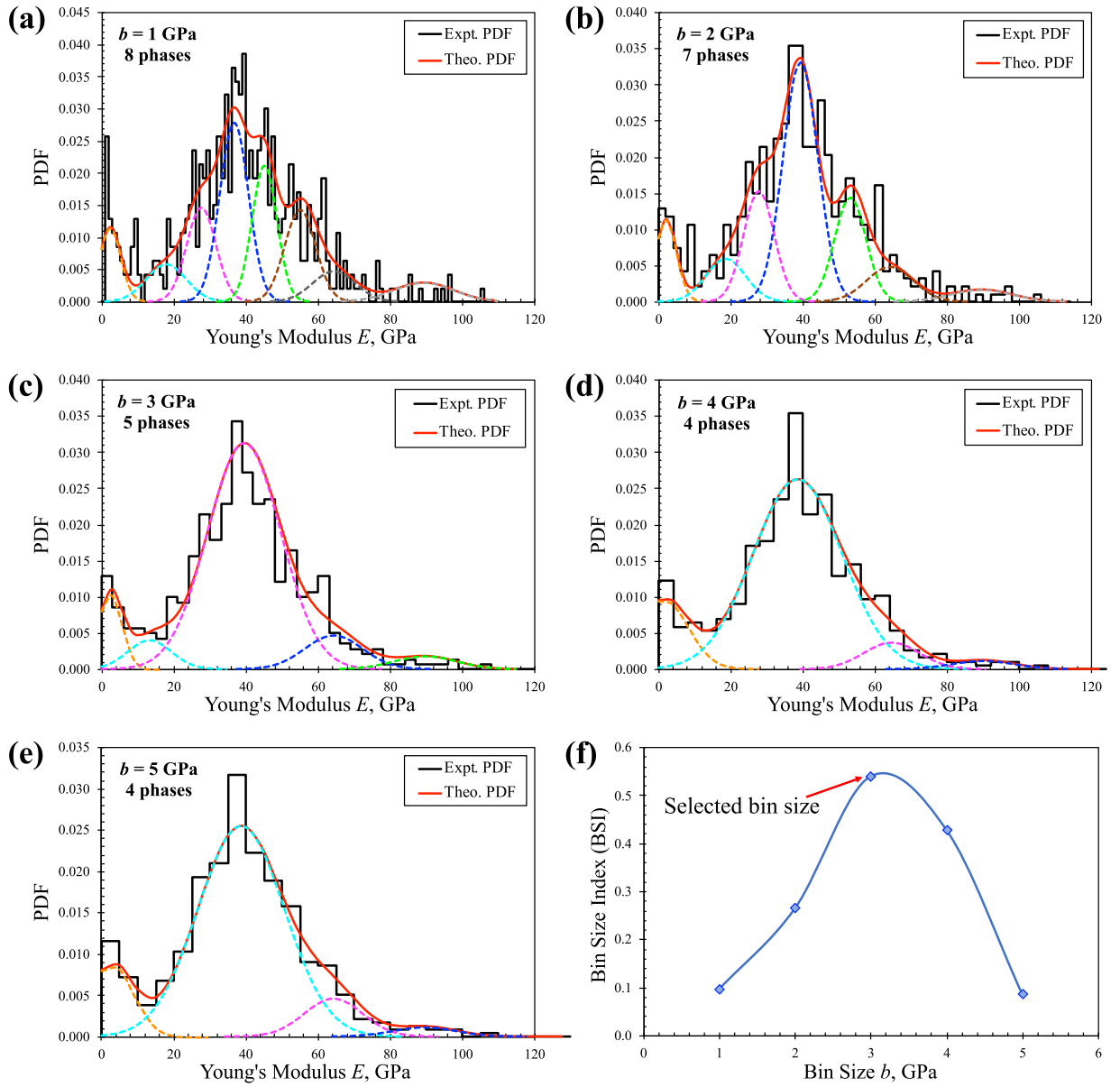


Fig. 5. Example results of deconvolution of the same sub-dataset on the Young's modulus of Sample A at the indentation depth of 0.5 μm showing the effect of variable bin sizes: (a–e) different number of phases resulting from different trial bin sizes; (f) the summary plot of the bin size index versus trial bin size.

each of the five trial bin sizes, a goodness of fit criterion called standard error (S_E) can be calculated after the least-squares fitting by following:

$$S_E = \sqrt{\frac{SSE}{DOF}} \tag{2}$$

where DOF is degree of freedom defined as the total number of input data points minus the total number of parameters used in the least-squares fitting, and SSE is sum of squares due to error, which is determined by:

$$SSE = \sum_{i=1}^{N} w_i(\hat{y}_i - y_i)^2 \tag{3}$$

where y_i and $\hat{y}_i$ are the ith of the total N input values and the predicted values, respectively, and w_i is the weight applied to each data point, usually $w_i = 1$.

Since the S_E values determined from different PDF deconvolutions can differ by more than one order of magnitude and are much smaller than 1 (usually at the order of 10^{-3}), they are further processed with a step of mean normalization to get the mean-normalized standard error S_{EN} whose values are scaled up to ~1:

$$S_{EN} = \frac{|S_E - \mu_S|}{\sigma_S} \tag{4}$$

where μ_s and σ_s are the mean and the standard deviation of the S_E from all fittings, respectively.

In this equation, the mean-normalized standard error S_{EN} is intentionally penalized or averaged by the number of the phases K. Inspired by the maximum likelihood-based criteria such as Akaike information criterion (AIC) and Bayesian information criterion (BIC), a new concept, Bin Size Index (i.e., BSI), is then proposed in this paper as follows by involving S_{EN} as well as the number of independent random variables K determined from each PDF-based deconvolution:

$$\mathrm{BSI} = \frac{|2 \times \ln(S_{EN})|}{K} \tag{5}$$

In general, the BSI exhibits a unimodal peak if it is plotted against the bin size b. As such, the optimal bin size is the one corresponding to the maximum BSI value. For the sub-dataset extracted at 0.5 μm segmentation depth, the optimal bin size is $b = 3.0$ GPa because it gives the highest BSI value among all five trials (Fig. 5). As mentioned previously, the measured E-h curves become less scattered with increasing indentation depth (Fig. 4b), and thus varied, instead of fixed, optimal bin sizes should be used for the sub-datasets extracted at different indentation depths to take into account of the variation in the actual range (i.e., the difference between the maximum and minimum values) of measurements with depth. Fig. 6a presents the dependence of range of Young's modulus measurements (ΔE) upon the indentation depth for Sample A, which offers a simple way to select optimal bin sizes for different sub-datasets. After the first optimal bin size is determined at an initial small depth of 0.5 μm, selecting optimal bin sizes for other sub-datasets can simply be based on the relationship between the ΔE-h relationship, i.e. the selected optimal bin size decreases at roughly the same rate as the reduction in the ΔE.

To further verify the performance of the BSI, the sub-dataset of Sample A extracted at the segmentation depth of 3.0 μm was analyzed in the same way as that described above. From Fig. 6a, at $h = 3.0$ μm, ΔE = ~60 GPa, and the optimal bin size is $b = 1.7$ GPa. This time, the Freedman-Diaconis rule (Freedman and Diaconis, 1981) suggests $b = 1.9$ GPa. Together with the bin size of 1.7 GPa determined from Fig. 6, a series of different trial bin sizes, 0.5, 1.0, 2.5, 3.5, and 4.5 GPa can then be examined, and the results are plotted in Fig. 7. As such, the optimal bin size is $b = 1.7$ GPa (Fig. 7f), which affirms that the bin sizes chosen from Fig. 6a are reasonable and universal for different sub-datasets. Therefore, it demonstrates that a varied bin size should be used for the construction of experimental PDFs of sub-datasets with different ranges of the measurement values from multiple random variables (i.e., the difference between the maximum and minimum values of all experimental measurements). Moreover, since Sample B has similar mineralogical compositions and measurement features (e.g., depth-dependent elasticity and range of measurements) as Sample A, a similar bin size selection plot is generated and used for the PDF-based deconvolution of the measurements obtained from Sample B (Fig. 6b).

In summary, the optimal bin size should be varied according to the range of the experimental measurements, and the BSI should reflect the tradeoff between the goodness of fitting (or standard error) and the number of fitted random variables. The optimal bin size is the one corresponding to a maximum BSI. For the studied shales, a simple way to choose varied optimal bin size is to follow the ΔE-h curves, provided that at least 1–2 optimal bin sizes are determined through the BSI criterion. Finally, the BSI criterion can expectedly be used for the analysis of other universal measurements of multiple random variables. However, whether or not the simple method for choosing optimal bin size can be applicable to other multiphase materials warrants further investigation.

3.6. Deconvolution results

Upon the selection of optimal bin sizes, the PDF histograms of all sub-datasets extracted at different segmentation depths were constructed. Fig. 8 shows the experimental PDFs and corresponding deconvolution results obtained from various segmentation depths for both Samples A and B. At smaller indentation depths, such as 1.0 μm, five phases with distinct mechanical properties are identified by deconvolution, and the overall PDF includes a large peak at the center and two smaller peaks on each side of the largest peak. By referencing the XRPD analysis as well as other published data on the elastic modulus of different minerals, the deconvoluted four small peaks on the two sides of the largest peak are assigned as quartz, carbonates, clay matrix, and organic matter (e.g., kerogen in voids), respectively, in the order of decreasing Young's modulus. However, caution should be taken in interpreting the largest peak that accounts for the highest fraction of the indentation measurements.

Prior microscopic observations on sample surfaces (Fig. 3) have revealed that the mechanically hard minerals (e.g., silicates, carbonates, and pyrite) are embedded in the clay matrix as solid inclusions. Based on published work that observed residual indents on sample surface (Yang et al., 2017), it is logical to assume that, for the randomly selected indentation locations, a certain number of indents are located right at or close to the boundaries between these stiffer solid inclusions and the surrounding softer clay matrix. Besides the visual evidences, this hypothesis is further confirmed by other similar work on shales involving coupled indentation and energy dispersive X-ray (EDX) spectroscopy analyses, which isolates interfaces between clay-rich and calcite-rich zones from other phases possessing different chemo-mechanical properties (Abedi et al., 2016).

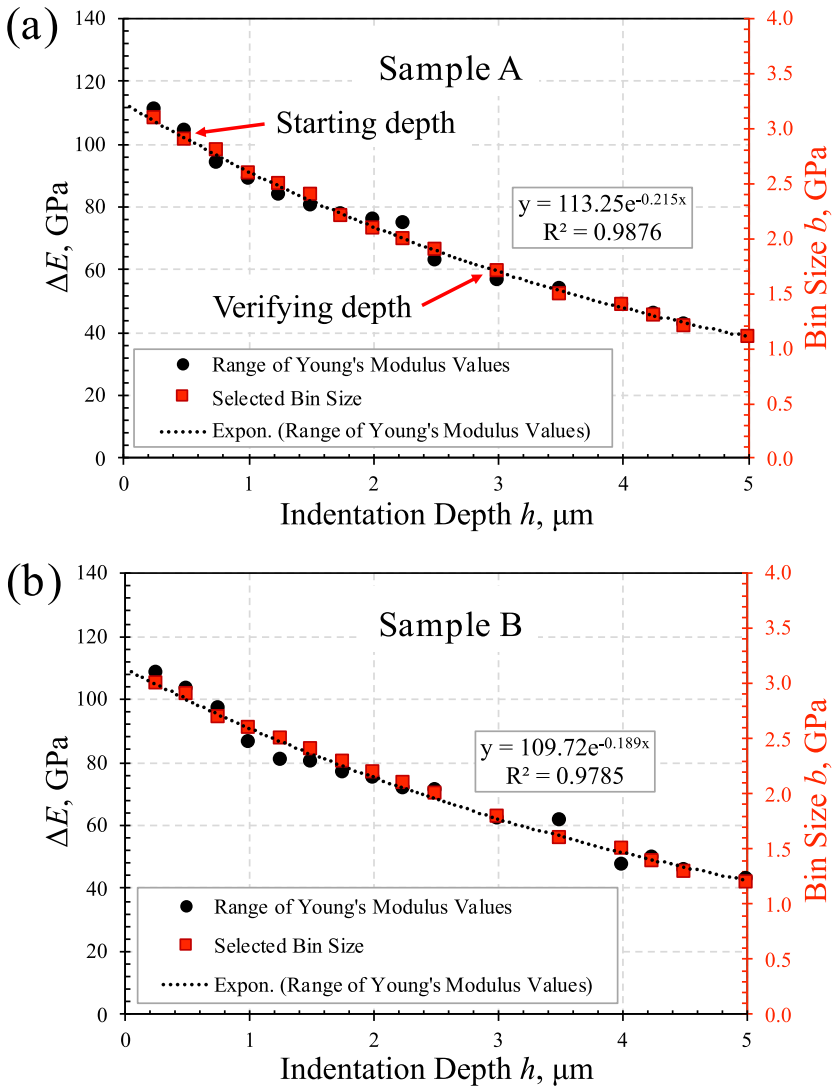


Fig. 6. The dependence of optimal bin size upon depth follows the same trend of ΔE-h curves: (a) results for Sample A; (b) results for Sample B (Note: in each figure, the scale of the two vertical axes are intentionally adjusted so that the two sets of data plots follow the same fitting equation).

From the statistical viewpoint, each indent is an independent statistical event in grid indentation. To approximately assess the probability of indenting on the interface for a two-dimensional binary composite consisting of a homogeneous matrix and solid inclusions as Phases 1 and 2 respectively, three imaginary cases depicted in Fig. 9 are considered. In all three cases, the area ratio of Phase 2 to Phase 1 is set to be 30% and remains unchanged. In Case 1, Phase 2 is simply a circular grain, but for Case 2 and Case 3, Phase 2 is replaced by two and three smaller circles, respectively, with a constant diameter in each case. Moreover, indentation measurements require certain contact between the indenter tip and sample surface, and hence the contact area is not infinitesimal. If the equivalent diameter c of the residual indent impression is approximate 15 μm (i.e., corresponding to indentation depth of ~2 μm for a Berkovich tip) and the interface zone is defined as the area of a ring formed by all possible residual indents that are in direct contact with the circular boundary of Phase 2 (i.e., both inside and outside the Phase 2, Fig. 9), then the probability of making an indent on the interface zone can be determined by dividing the area of the interface ring by the total area occupied by the composite (i.e., area of the square). The probabilities of indents located on the interface zone for the three cases are 5.1%, 7.2%, and 8.8%, respectively, which are actually conservative because (1) the variable sizes of the inclusions (i.e., mostly grains are actually much smaller) are not considered in these cases, and (2) the size of the elasto-plastic zone beneath the indenter tip usually exceeds the actual contact area or residual impression size (i.e., purely plastic deformation). Therefore, due to the fact that shale is a highly heterogeneous composite with numerous small inclusions, grid indentation can result in a high probability of detecting the interface between hard and soft minerals, which is a virtual, but not a real, phase extracted from statistical deconvolution.

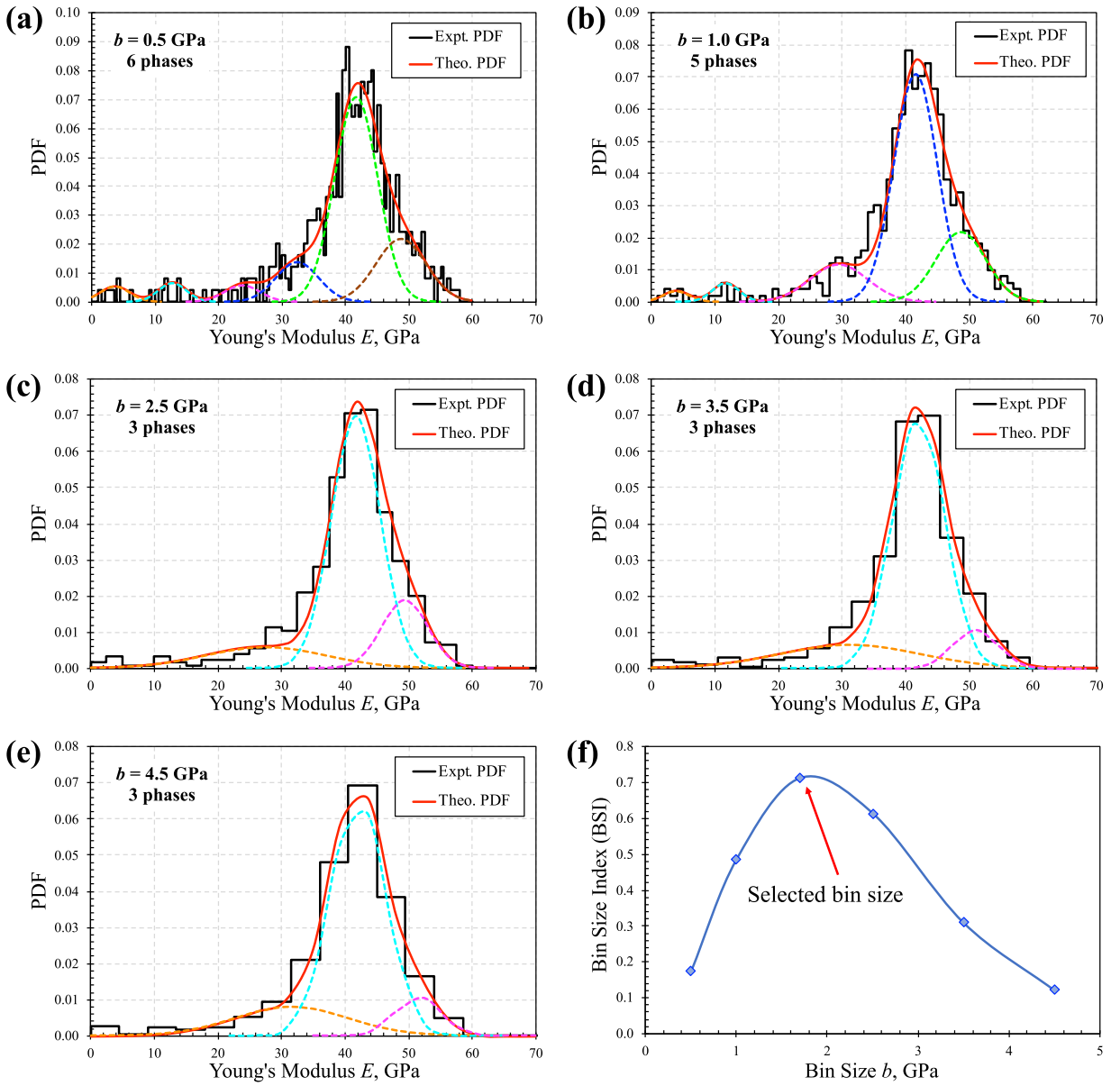


Fig. 7. Example results of deconvolution of the same sub-dataset on the Young's moduli of Sample A at the indentation depth of 3.0 μm showing the effect of variable bin sizes: (a–e) different number of phases resulting from different trial bin sizes; (f) the summary plot of the bin size index versus trial bin size. Note that the results for the optimal bin size b = 1.7 GPa is provided in Fig. 8c.

For each segmentation depth, by extracting the mean of each Gaussian distribution (i.e., each deconvoluted phase at the same depth) and plotting these means against the segmentation depth at both linear and semi-logarithmic scales, the Young's modulus-depth curves, which seem disorganized and irregular, are classified into five different groups, assigned as quartz, carbonates, interface, clay matrix, and organic matter (Fig. 10). Noteworthy is that these lines are generated by a special criterion: since the virtual interface phase has the highest PDF frequency as well as the greatest fraction, the mean values obtained from each deconvoluted sub-dataset are linked by always maintaining the largest peak as the interface. In other words, the deconvoluted, highest peaks from all sub-datasets (i.e., at all segmentation depths) are linked as one curve to represent the virtual interface phase. Then the peaks on the left side of the interface phase are linked to generate the curves for softer phases (i.e., clay matrix and organic matter), while those on the right side are linked for harder phases such as carbonates and quartz.

The total number of phases presented in Fig. 10 is less than that identified by XRPD analysis, because of three reasons: (1) some minerals only exist at very small amounts (e.g., 2–3%), and hence may be omitted statistically; (2) the particle sizes of

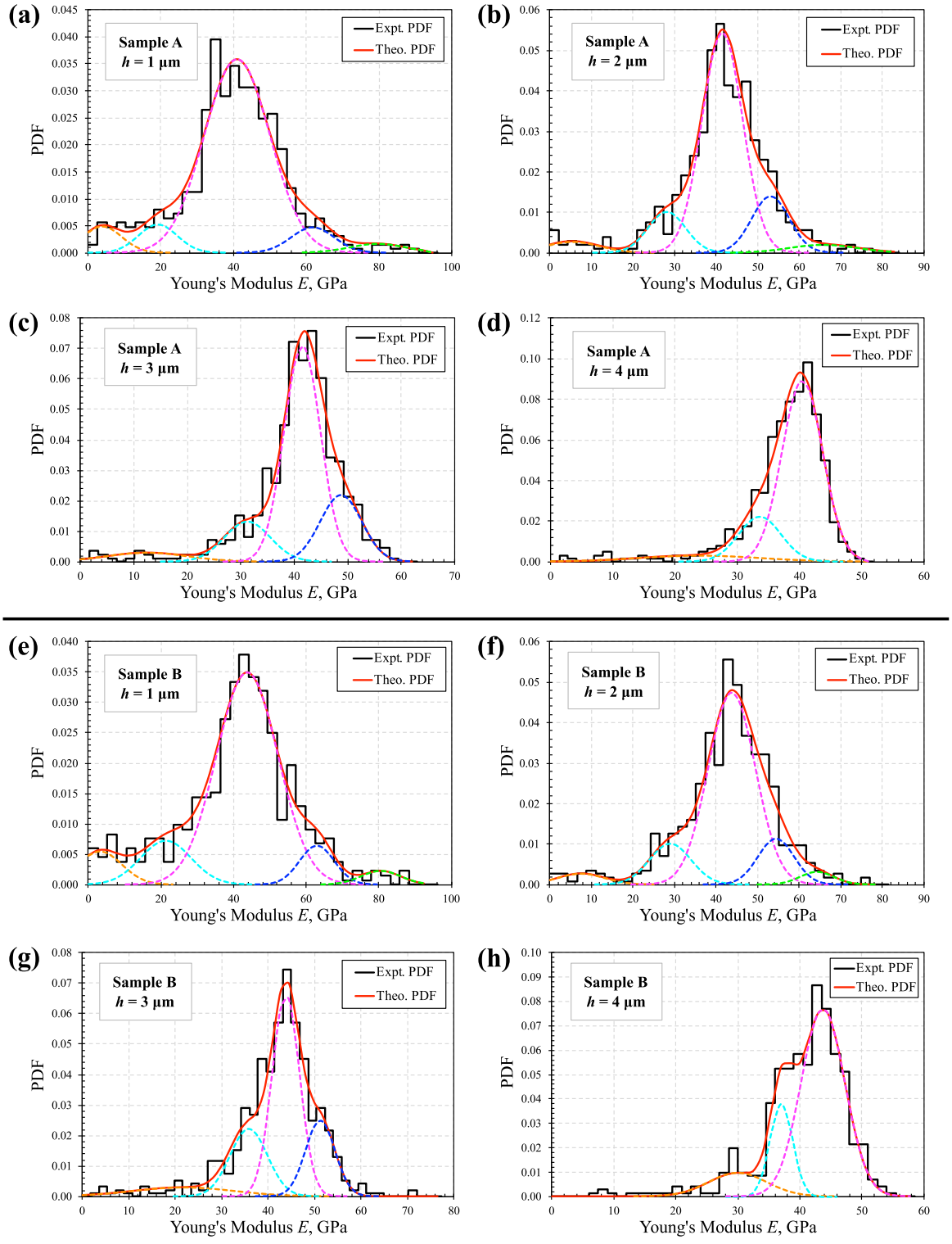


Fig. 8. Representative experimental PDFs and corresponding deconvolution results for Samples A (a–d) and B (e–h) at indentation depths of 1, 2, 3, and 4 μm, respectively.

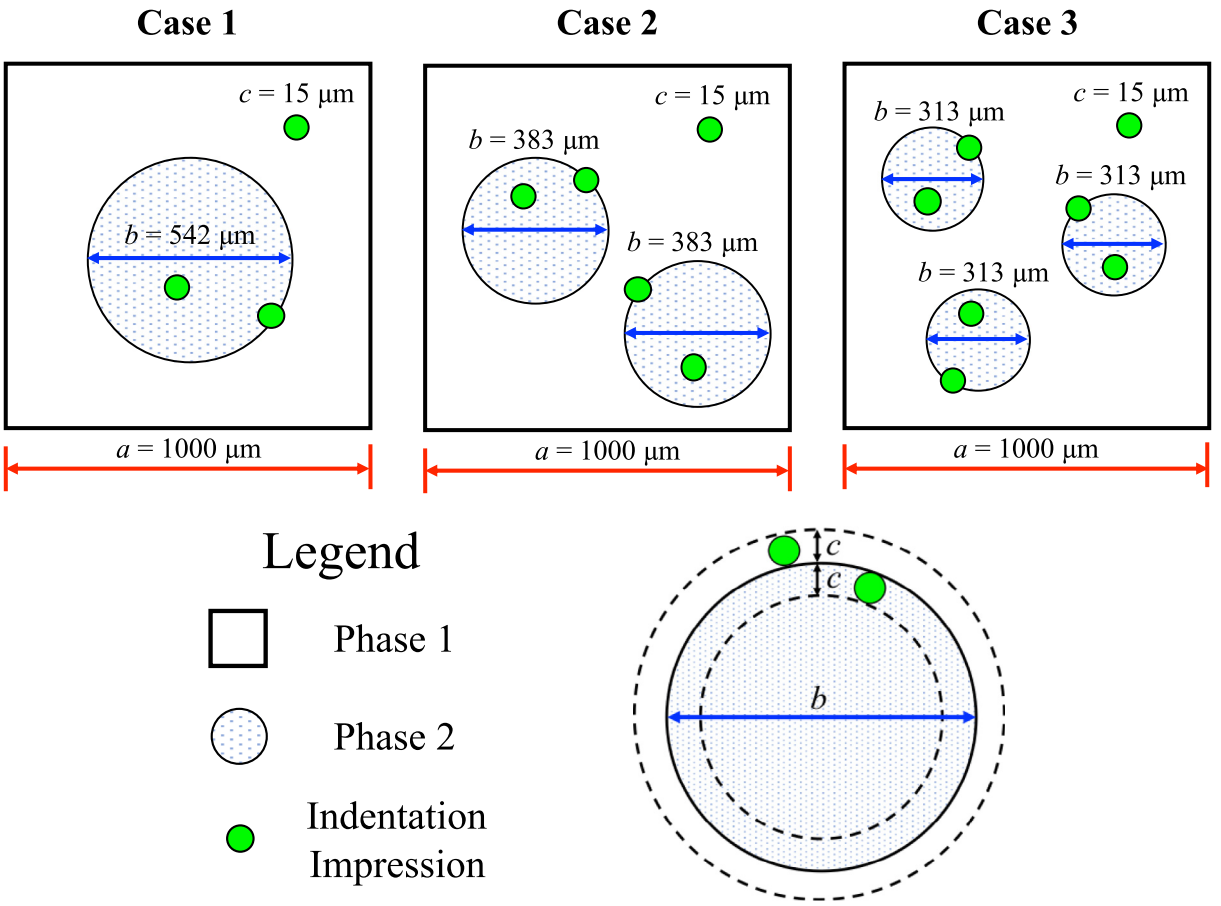


Fig. 9. Three simplified cases illustrating the possibility of indenting to a finite depth at the interface zone on an imaginary binary composite (not to scale).

some inclusions may be extremely small, and hence they are included and mixed within the clay matrix; (3) it may be likely that two or more phases have similar mechanical properties that cannot be distinguished statistically by deconvolution, and hence these mechanically similar phases are grouped into one phase by deconvolution. In other words, insufficient contrast of mechanical properties can also render the separation of different phases impossible in the deconvolution analysis. In essence, this may be a significant limitation of nanoindentation-based phase identification and quantification.

As shown in Fig. 10, the Young's modulus of the virtual interface phase almost remains constant with indentation depth, while those of the other four phases exhibit significant depth-dependent behavior. For instance, the Young's moduli of organic matter and clay matrix increase with depth, while quartz and calcite show the opposite trend. Meanwhile, the number of the deconvoluted peaks gradually decreases with depth, and at the depth of 5 μm, only one unique peak exists. This intuitively demonstrates that if the indentation depth and hence the size of the elasto-plastic zone beneath the indenter is sufficiently large, the measured mechanical response is the behavior of the bulk rock (i.e., as a composite). Moreover, the effect of indentation depth on the mechanical behavior depends upon the relative stiffness and hardness of different phases within the rock. For softer phases, the nominal deconvoluted Young's modulus increases with depth, while the stiffer phases show the opposite trend.

Also shown in this figure are the Young's modulus of the interface phase, averaged from the 20 indents selectively made exactly on the interface under the assistance of the optical microscope, as well as the ex-situ indentation results of relatively large quartz and carbonate particles. It can be clearly observed that the selectively-measured Young's modulus of the interface phase (i.e., the one labeled as "Interface_selected") is consistent with the one from statistical deconvolution for Sample A, except the data at 0.25 μm depth, which may be caused by surface roughness effect. For both samples, the Young's moduli of the deconvoluted carbonate phase are consistent with that of large pure carbonate grains, but the Young's moduli of the deconvoluted quartz phase are slightly lower than the ex-situ measurements obtained from large quartz grains, which is likely caused by the surround effect (i.e., the relatively lower modulus of clay matrix surrounding quartz grains), which is further discussed later.

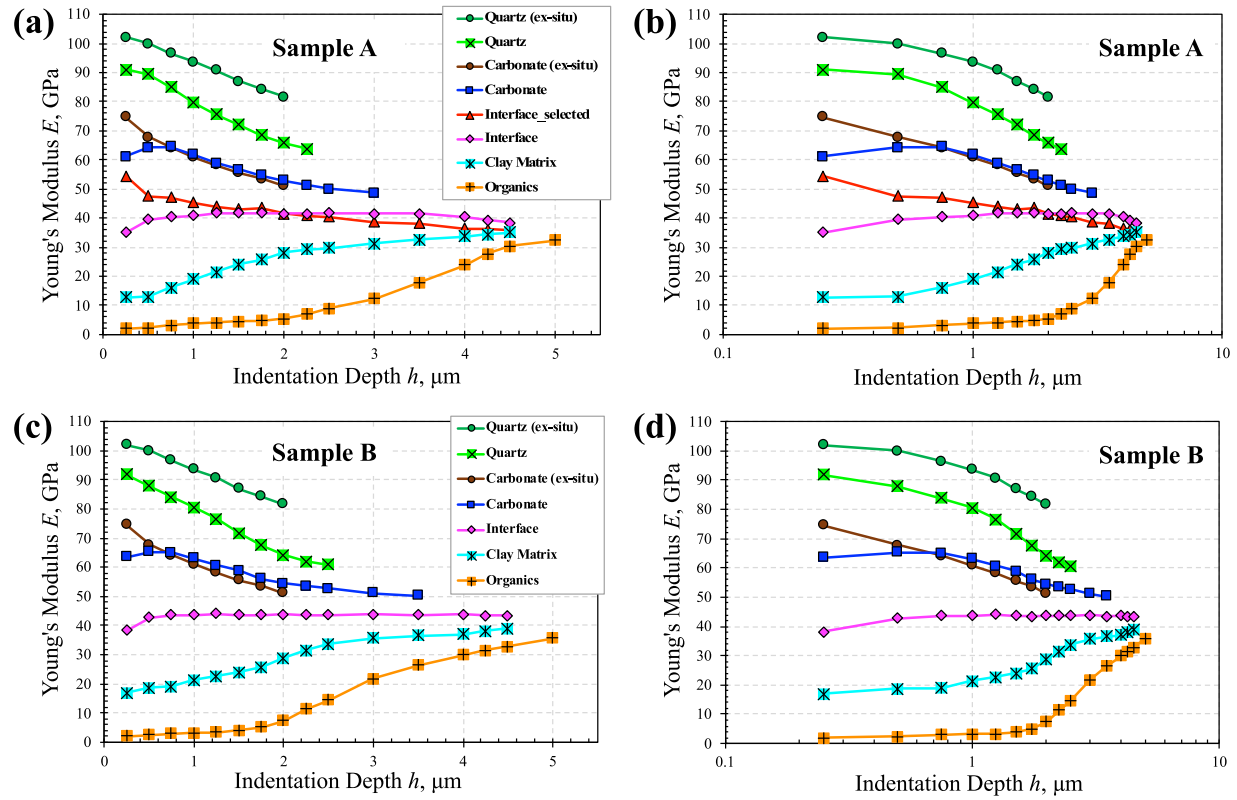


Fig. 10. Plots of deconvoluted Young's modulus of individual phases versus indentation depth at both linear scale (left) and semi-logarithmic scale (right): (a–b) Samples A; (c–d) Sample B.

3.7. Thin film-substrate analogy

The size-dependent elasticity has been reportedly observed in studying the mechanical behavior of nanowires and nanobeams (Agrawal et al., 2008; Yao et al., 2013; Zheng et al., 2014). In this study, a variety of factors may play a role in the dependence of elasticity on indentation depth. Fig. 11 shows typical SEM images of residual indents selected from the grid indentation, and their corresponding indentation data manifest that the dimensions and spatial distribution of different mineral phases have direct influence on the observed behavior. For the indent made on a single phase with sufficiently large in-plane dimensions (Fig. 11a), the loading curve is relatively smooth, and the measured Young's modulus decreases slightly with indentation depth. However, as shown in Fig. 11c and d, steps or serrations are observed on the loading curve. These steps occurring on the *F-h* curves are known as 'pop-ins' where the indentation depth continuously increases while there is little change in load (Zhang et al., 2009). Many physical and mechanical events may contribute to the occurrence of pop-ins, it is known that the material's heterogeneity will contribute to the pop-in behavior as the elasto-plastic zone moves across the boundary of two phases. In Fig. 11c, since the initially indented phase only has dimensions of ~10 × 5 μm, cracks occur when the indentation depth or contact area is large enough to interact with the surrounding phases, which appear as the pop-ins on the loading curves and result in significant reduction in the measured Young's modulus. Another significance is that, while the Young's modulus in Fig. 11c decreases continuously with depth, both significant decrease and increase (or large and small "V"-shaped sections) are observed in Fig. 11d. Noteworthy is that the indent in Fig. 11a seems to be made in a single phase zone, while that in Fig. 11c is on a multiphase zone with pores or cavities. As such, when the elastic zone expands from a softer phase into a harder phase, the overall mechanical response is then stiffer, leading to an increase in the measured Young's modulus.

For a naturally occurring material such as shale with high heterogeneity, it is very difficult in practice to assess the spatial distributions of the constituent phases beneath each indent before and after performing grid indentation testing. Finite element analysis of nanoindentation (Fischer-Cripps, 2007) reveals that the elastic zone within the tested material induced by the indenter tip under indentation loading is significantly greater than the size of the residual indent (i.e., the fully developed plastic zone). For the indents initially located at the interface region of two adjacent phases, the measured Young's modulus of the virtual interface phase already reflects the combined property of two different phases. As such, it remains relatively constant until large indentation depths where more extensive cracking has occurred. On the other hand, even though the designated indent originally locates on a single phase, as shown schematically in Fig. 12, the elastic

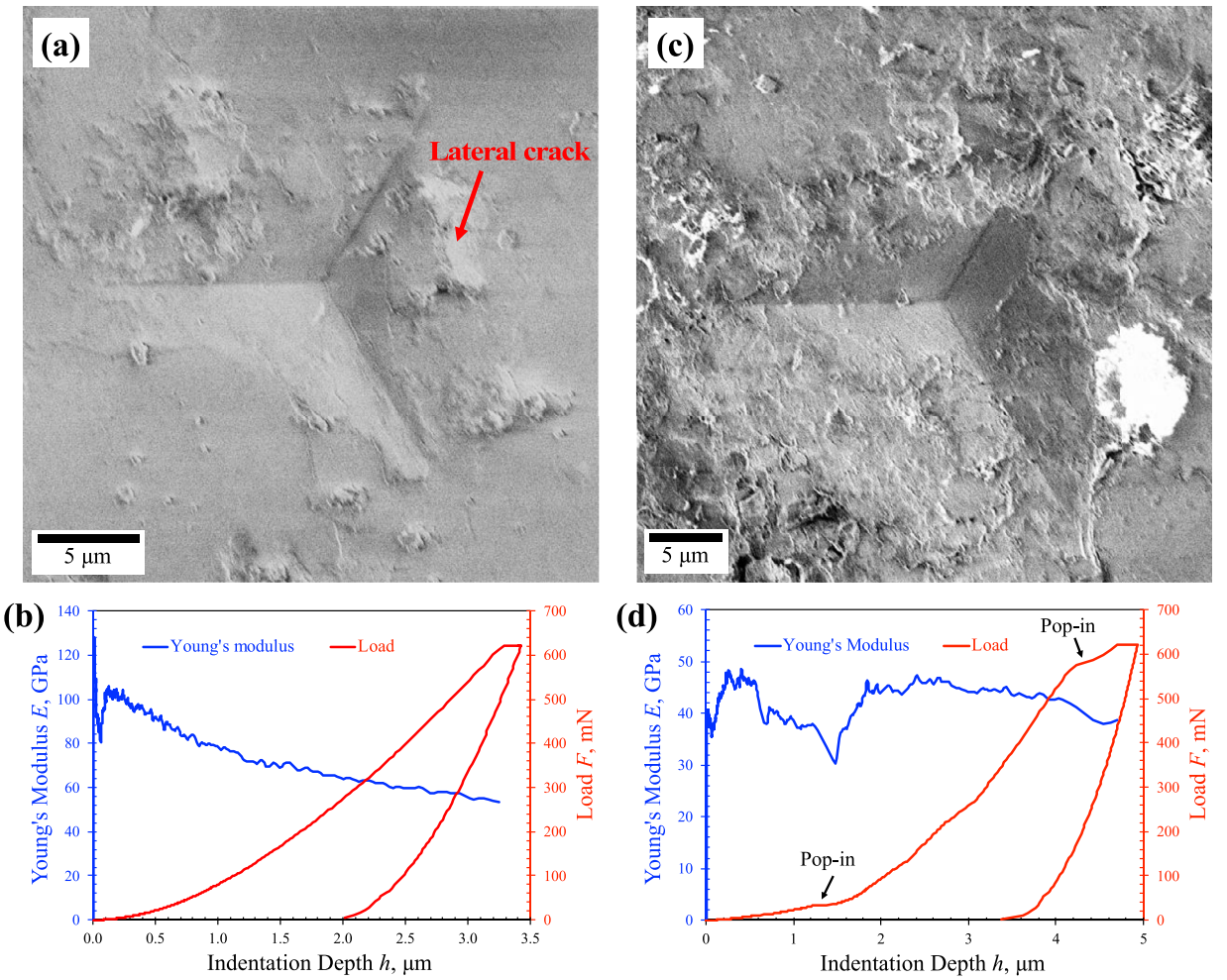


Fig. 11. SEM micrographs of selected residual indents and their corresponding indentation measurements: (a–b) an indent made on a large particle; (c–d) an indent made on a multiphase (i.e., clay matrix) zone.

zone can still extend from the initially indented phase to other surrounding phases when the entire elastic zone expands with increasing indentation depth beyond the dimension of the indented phase, resulting in changes (i.e., either increase or decrease) in the measured Young's modulus over depth. In addition, for composites possessing a high ratio of Young's modulus between hard particles and soft matrix, numerical simulation results show that a whole stiff particle is likely to sink into the surrounding soft matrix and stops deforming further as the indenter proceeds, which can result in a significant underestimation of the elasticity at greater indentation depths (Leisen et al., 2012).

The shapes of the *E-h* curves in the previous semi-logarithmic plots (Fig. 10b and d) are similar to the results obtained from the thin film-substrate systems, including a soft film on a hard substrate and a hard film on a soft substrate (Fig. 12 inset) (Wei et al., 2009). Similarly, finite element simulations demonstrate that for a two-phase material with a yield strength ratio of 0.5–2, the thin film-substrate geometry can yield much severer restriction on the depth of indentation in the determination of individual phase's properties when compared with other possible spatial arrangements, such as a square particle embedded in a semi-infinite medium and a semi-infinite fiber embedded in a semi-finite medium (Durst et al., 2004). Therefore, the thin film-substrate system is usually used as an analogy to heterogeneous materials in order to reduce the complexity of analysis of the mechanical response from a multiphase material (Constantinides et al., 2006). In general, a conservative rule of thumb, 10% of the maximum indentation depth, is generally adopted to avoid the significant influence of the underlying substrate on the measurements of thin film properties. That is, the maximum indentation depth should not exceed 10% of the film thickness in indentation testing, because the indentation depth is empirically estimated to be about 1/10 of the size of the interaction volume beneath the indenter, and nanoindentation testing measures the average response of all phases inside this volume. Nevertheless, this so-called 10% rule is not necessarily true for the measurements of elastic modulus of certain materials or composites (Bückle, 1973; Fischer-Cripps, 2011; Saha and Nix, 2002).

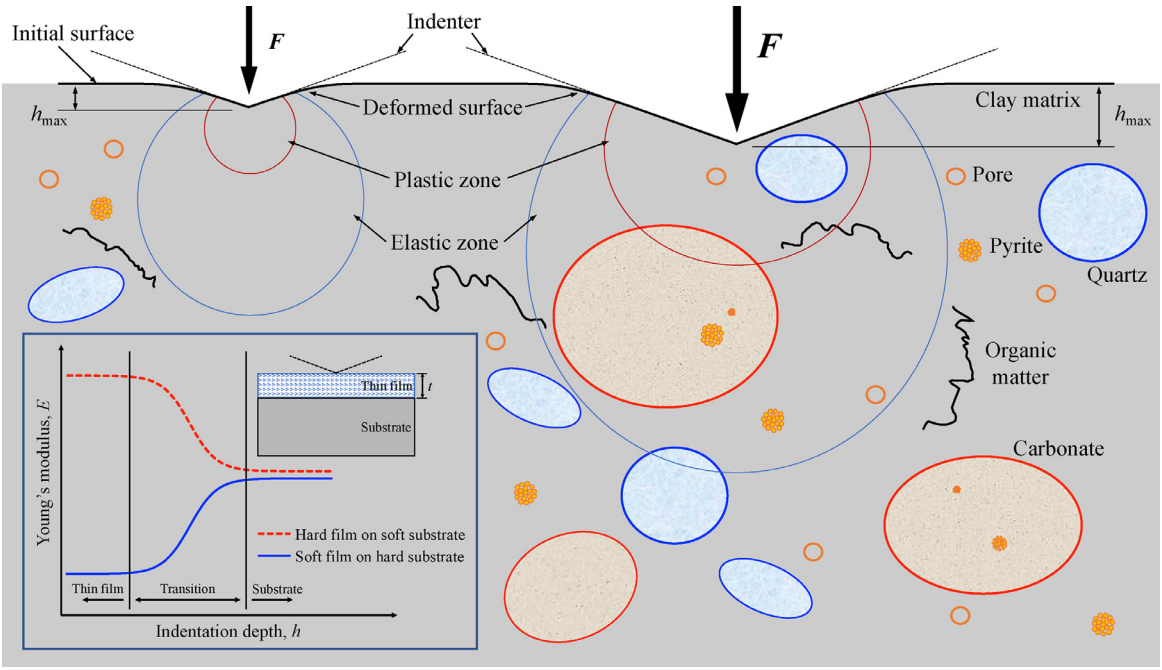


Fig. 12. Schematic illustration of the surround effect occurring in nanoindentation performed on shale. The inset shows the thin film-substrate analogy to this effect.

3.8. A surround effect model

The multiscale mechanical properties of an organic-rich shale were assessed by Bennett et al. (2015) via conventional nanoindentation experiments performed at four length scales of indentation depth ranging from 0.2 to 5 μm, with the purpose of extracting the mechanical properties of different phases (e.g., quartz, clay matrix) at different indentation depths. In this study, the model originally developed for the thin film-substrate system (Wei et al., 2009) is referenced but revised to describe the cross-scale elasticity of shales probed by nanoindentation testing, i.e., the interesting phenomenon of depth-dependent elasticity caused by the surround effect. Here, the deconvoluted Young's moduli of each phase presented in Fig. 10, except for the virtual interface phase, are fitted by the following model:

$$E = E_c + \frac{E_p - E_c}{1 + [h/(t\alpha)]^Y} \tag{6}$$

where t is the characteristic length scale of the individual phases; Y and α are two constants determined through the model fitting; E_p and E_c are the Young's moduli of individual phases and the bulk composite, respectively; E and h are the input data where E is the Young's modulus of the fitted phase deconvoluted at a specific indentation depth of h. The model is fitted by performing nonlinear regression in a statistical analysis platform R using the Levenberg-Marquardt iterative algorithm (Levenberg, 1944; Marquardt, 1963), and the $t\alpha$ term is treated as a whole since the characteristic length scale (i.e., the "film thickness" t) of each phase is unknown. For the two relatively soft phases, clay matrix and organic matter, all parameters were fitted without any constraints. Nevertheless, as previously mentioned, indentation data of stiffer phases (e.g., quartz and carbonate) are not available at relatively large indentation depths due to the limited maximum load capacity of the nanoindenter instrument. As such, the parameter E_c (i.e., the Young's modulus of bulk composite) cannot be fitted within the maximum number of iterations for these two phases unless a lower bound of E_c is specified. In this case, the E_c determined from the fitting of the clay matrix phase was set to be the lower bound for fitting the quartz and carbonate phases, and such restriction will not influence the fitting of the other three parameters in the model.

Fig. 13 shows the plots of Young's modulus versus indentation depth, including the experimental measurements presented in Fig. 10 and pertinent curves of the surround effect model obtained via nonlinear regression. The surround effect model yields a continuous (i.e., cross-scale) range of the Young's modulus of various mineral phases with indentation depth, and a gradual transition in Young's modulus is observed from the individual phase-dominated response to the bulk material-dominated one. For example, the representative Young's moduli of quartz, carbonate, clay matrix, and organic matter in Sample A are determined to be 89.7, 63.0, 12.0, and 3.1 GPa, respectively, at very small depths, while the Young's modulus of the bulk rock is within a very small range of 35.3–45.3 GPa, determined from depths of >~10 μm. The discrepancy in the Young's modulus of the bulk rock is due to the unconstrained regression for the clay matrix phase and the organic phase

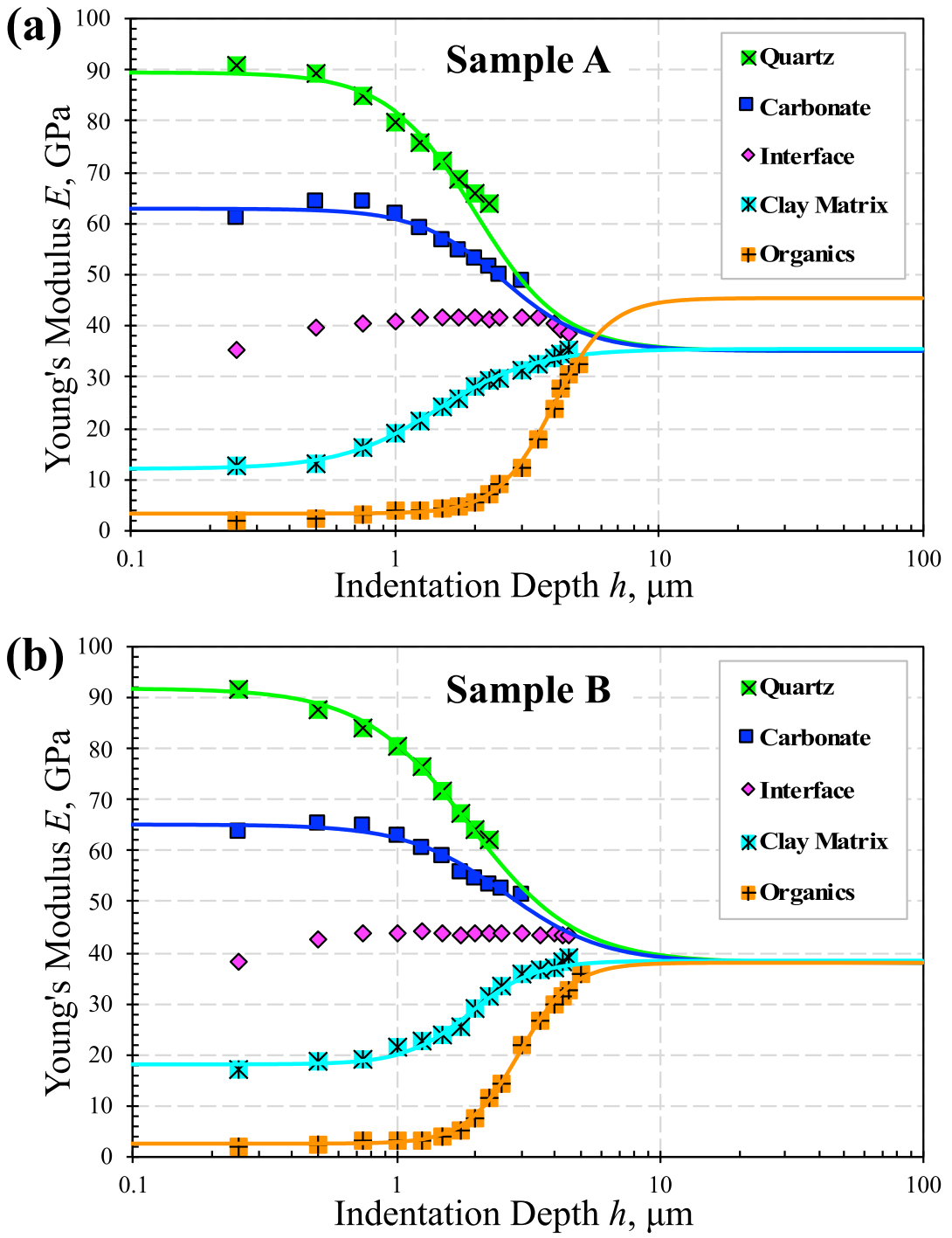


Fig. 13. Nonlinear regression of the Young's modulus versus indentation depth curves using the surround effect model: (a) Sample A; (b) Sample B.

as well as to the lack of data point at large depths (again, resulting from the limitation of the instrument's maximum load capacity). However, it seems not an issue for Sample B.

Further noteworthy are other interesting observations on the surround effect model-fitting curves. As a nature of the model, the fitting curves approach asymptotically to the E_p and E_c at zero or infinitesimal and infinitely large depths, respectively. Although it is theoretically possible, obtaining accurate indentation measurements at very small and very large depths is practically very challenging. The former is usually affected by surface roughness, surface contamination, indenter tip bluntness, and indenter tip area function, while the latter is complexed by cracking (that tends to occur under large

Table 3
Summary of results of the two samples obtained from the newly proposed data analytics and micromechanical models.

Properties	Organics	Clay matrix	Carbonate	Quartz	Bulk rock	VRH average	MT model
Sample A							
Mean particle size (μm)	11	4	10	5	90		
Young's modulus (GPa)	3.1	12.0	63.0	89.7	35.3–45.3	38.6	32.8
Fraction (vol.%)	19	28	38	15			
Fraction by XRPD (wt.%)		37	30	20			
Sample B							
Mean particle size (μm)	10	7	7	5	110		
Young's modulus (GPa)	2.3	18.2	65.1	92.0	37.9–38.6	37.7	33.8
Fraction (vol.%)	22	39	26	13			
Fraction by XRPD (wt.%)		38	27	18			

load), sample size, and instrument's maximum load capacity, among others. However, with the new surround effect model, these detrimental factors, even if unavoidable, can be overcome, leading to more accurate determination of the elasticity of both individual phases and the bulk rock. Moreover, the first turning point (i.e., the one at shallower depths) on each fitting curve is the critical depth or maximum allowable indentation depth for characterizing individual phases' properties, while the second turning point at larger depths is the critical depth for the indentation measurement of the properties of the bulk rock. These two points can actually guide the selection of appropriate depths for the measurement of elasticity of each phase.

With the new model, quantitative analysis of the volumetric fractions of each phase becomes feasible. They can be calculated by averaging the fractions determined from different deconvolution analyses conducted at the segmented depths smaller than the critical depth. Noteworthy is that the actual volumetric fraction should be the recalculated percentage by discarding the fraction of the virtual interface phase. The results obtained from the two investigated shale samples are summarized in Table 3. Since the specific gravities of these common rock-forming minerals, except pyrite that is present at a very small fraction, only vary in a narrow range (e.g., 2.65–2.80), the volumetric fractions should be approximately the same as the weight-based fractions. In addition, also shown in the table are the weight-based fractions of all mineral phases determined by quantitative XRPD analysis. In fact, results from these two different independent approaches are comparable or consistent, validating that the data obtained from the newly proposed data analytics can also be used to analyze the volumetric fractions of the mechanically distinct phases.

In rock physics and materials science, various micromechanical models have been developed for theoretical prediction of bulk rock's (or composites) elastic modulus based on the known volumetric fractions and moduli of individual constituents, which involves the process of homogenization and upscaling from the smaller scale to the larger scale. In this paper, two commonly used methods, Voigt-Reuss-Hill (VRH) average (Mavko et al., 2009) and Mori-Tanaka (MT) model (Benveniste, 1987; Huang et al., 2014; Mori and Tanaka, 1973), were examined to validate the experimental results obtained from the nanoindentation. The MT model is generally applicable to composites with a matrix-inclusion microstructure, such as shales, where quartz and carbonate are considered as the solid inclusions within the clay matrix. In this study, the volumetric fractions of different individual phases can be determined by the data analytics, and results are rescaled to a total of 100%. The upscaled Young's moduli of the bulk samples calculated based on the VRH and MT models are also summarized in Table 3. A simple comparison finds that the macroscale elastic moduli of two shales predicted by both models agree well with those determined by the data analytics. Moreover, the latter approach shows additional advantages: cross-scale characterization of the Young's modulus is achieved via only a single experimental approach, and hence the homogenization process from individual constituents to bulk rock can be visualized and explicitly uncovered.

Furthermore, the characteristic size of each mineral grains can also be indirectly estimated by the critical depth using the 10% rule of thumb, and results are summarized in Table 3. These sizes are nominally the average characteristic length scale or the mean particle size for each mechanically distinct phase detected by nanoindentation, and the actual size of each mineral grain should vary within a range. Noteworthy is the bulk rock's characteristic length scale, which is also the minimum size of the elastic zone required to obtain the composite elastic response of the bulk rock. This size is actually 10 times greater than the actual minimum indentation depth required to measure the elastic modulus of the bulk rock.

As expected, the critical depths shown in Fig. 13 are not the same for different phases in each tested sample, which confirms that assuming only one characteristic size or using one fixed indentation depth for all different mineral phases in conventional grid indentation technique is far from adequate for accurate determination of mechanical properties of individual phases in heterogeneous, multiscale materials such as shales. Again, although the mechanical properties of individual phases can theoretically be probed at indentation depth much smaller than their characteristic sizes, but the quality of the finished surface (i.e., the surface roughness), in turn, significantly affects the accuracy of indentation measurements at such shallow depths (de Souza et al., 2006). Fig. 13 shows an example of the roughness effect in which the Young's modulus of most phases obtained from the deconvolution analysis at the depth of 0.25 μm is smaller than that determined by the surround effect model. The reported experimental results on bones (Donnelly et al., 2006) suggest that the effect of surface roughness on the indentation measurements can be avoided when the indentation depth is at least three times greater than

the RMS roughness of the tested surface. On the other hand, if the pre-selected indentation depth is too large (e.g., 1 μm as shown in Fig. 13), then the results obtained by conventional deconvolution analysis but without the surround effect model are significantly affected by the too large elastic zones that actually include surrounding minerals with dissimilar mechanical properties. Therefore, for conventional grid indentation (but without the surround effect modeling), the pre-selected maximum indentation depth needs to be properly balanced between the roughness effect and surround effect when probing individual phases' mechanical properties, which is typically challenging for multiphase and multiscale materials such as shales.

3.9. Summary of the new data analytics

As formulated above, the newly developed data analytics consists of the following major steps: (1) data segmentation to extract multiple sub-datasets at various segmentation depths from a large number (e.g., >500) of continuous *E*-*h* curves obtained from unknown phases or locations; (2) selection of an appropriate bin size and construction of experimental PDF for each sub-dataset, followed by statistical deconvolution of the PDF to determine the number of mechanically distinct phases and their Young's modulus at each segmentation depth; (3) data re-assembly by replotting the Young's modulus of each individual phase against segmentation depth to re-gain the depth-dependent behavior; (4) surround effect modeling by fitting the data with the newly proposed surround effect model; and (5) extraction and determination of the Young's moduli of both individual phases at the micro/nano-scale and the bulk material at the macroscale. In this study, applying the newly developed data analytics to processing the >500 highly scattered, chaotic *E*-*h* curves (Fig. 4) results in only a few highly informative curves that reflect the depth-dependent transition of the Young's modulus from different individual phases to the macroscale bulk rock. While the former, though containing voluminous data, yields little information about the shale as a multiphase, multiscale composite, extensive information can be obtained from the latter, which is summarized as follows:

1) The number, area or volumetric fraction, and Young's modulus of mechanically distinct phases in the shale;
2) Young's modulus of the bulk shale;
3) Appropriate indentation depths at which the Young's modulus of each mechanically distinct phase as well as the bulk rock can be more accurately measured;
4) Average particle size of each different phase in the rock.

In addition, other advantages of the data analytics include: (1) the mechanical properties of individual phases can be determined without the need for assuming characteristic lengths or sizes of individual phases; (2) the model-fitted results are less or not at all influenced by the surface roughness, especially for cases where some phases may have particle sizes less than or nearly equal to the roughness; (3) the mechanical properties of both individual phases and the bulk rock are determined simultaneously by one single experimental technique; (4) it validates that the measured mechanical properties depends upon indentation depth, and the bulk material's properties can be obtained when the indentation depth is sufficiently large to induce a large elasto-plastic zone encompassing all constituent phases; and (5) the bulk material's properties can be obtained without the need to make very large indents, which may be also difficult especially for hard and stiff samples, because such measurements require a high load capacity and special indenter tip with large effective depths, and a large indentation depth is usually accompanied with cracking, pileup or sink-ins that in turn affect the accurate determination of the contact area.

Another significant development is the formulation of a newly proposed concept, bin size index (BSI), and its application for selecting an optimal bin size that varies with indentation depth. Although it is developed for the analysis of the Young's modulus of shales, this criterion appears powerful for and is expectedly applicable to experimental measurements from a group of mixed independent random variables (i.e., not a single variable) but without the knowledge of the exact number of variables. Moreover, the criterion demonstrates that the optimal bin size can or should be varied to process different sub-datasets obtained at different conditions (e.g., different indentation depths in this study). Coincidently, the Young's modulus of the bulk rock is nearly the same as that of the virtual interface phase between hard and soft constituents, suggesting that the macroscopic properties of the shale may be estimated from the indents made on the interface of two mineral phases with contrasting mechanical properties. Finally, this technique can provide guides for selecting appropriate indentation depths to probe the mechanical properties of either highly heterogeneous bulk materials at the macroscale or their individual constituent phases at the microscale.

4. Conclusions

An improved statistical nanoindentation technique with novel data analytics are developed to cross-scale characterize the elastic properties of two shale samples. For highly heterogeneous, multiscale composites (e.g., shales) containing solid inclusions, this new data analytics, including data segmentation, bin size optimization, statistical deconvolution, data re-assembly, surround effect modeling, and property extraction, provides an innovative framework for converting voluminous, chaotic, and highly scattered data into highly concise, informative, and insightful results that are explicitly linked to the composite's cross-scale elasticity as well as certain microstructural features (e.g., the number of phases, sizes of solid inclusions). As discussed above, the following conclusions can be drawn:

- Cross-scale characterization of heterogeneous, multiscale, multiphase composites (e.g. shales) consisting of a homogeneous fine-grained matrix with embedded coarse-grained solid inclusions can be achieved by the improved statistical nanoindentation with the new data analytics;
- A new criterion, termed Bin Size Index, proposed to optimize depth-dependent, variable bin sizes for experimental PDF construction, was developed and validated using the massive nanoindentation data;
- A new surround effect model is first developed to account for the combined mechanical responses from mechanically dissimilar constituents in highly heterogeneous, multiscale, multiphase composites, and to obtain multifaceted results including the cross-scale elasticity of the composites as well as the microstructural features such as the number of mechanically distinct phases and their characteristic sizes.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Shengmin Luo: Investigation, Data curation, Writing - original draft. **Yunhu Lu:** Conceptualization, Data curation, Writing - review & editing, Funding acquisition. **Yongkang Wu:** Investigation, Data curation, Software, Validation. **Jinliang Song:** Software, Formal analysis. **Don J. DeGroot:** Supervision, Project administration. **Yan Jin:** Supervision, Resources, Funding acquisition. **Guoping Zhang:** Project administration, Writing - review & editing, Supervision, Conceptualization, Data curation, Funding acquisition, Methodology, Validation, Formal analysis.

Acknowledgments

This work was partially supported by the National Natural Science Foundation of China (Grant No. 51774305) and Major Projects of the National Natural Science Foundation of China (Grant No. 51490651). The authors are very grateful to Dr. Yibing Deng of the University of Massachusetts Amherst for conducting nanoindentation measurements on ex-situ quartz and carbonate sands. This work made use of the Shared Experimental Facilities (i.e., X-ray diffractometer) at MIT, supported in part by the MRSEC Program of the National Science Foundation under award number DMR1419807.

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